

Shivam Kumar

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SUMMARY

SDE II at Microsoft building Azure cloud-native platforms, distributed systems, and Kubernetes-based container infrastructure. Expertise in microservices architecture, AI/LLM-powered SRE automation, and Dapr (CNCF Graduated). IIT Kanpur, 5 YoE.

EXPERIENCE

Software Development Engineer II

Jul 2024 – Present

Microsoft – Azure Container Apps

Hyderabad, India

- Engineered AI-powered SRE agent for Azure Container Apps, reducing P50 TTM by **97%** and TTR by **64%**; added RCA subagents, self-repair, auto-resolution, and Ev2 deployment pipelines.
- Enhanced Azure Functions on ACA with support for graceful drain, secure key management, KEDA scale-rule customization, JWT clock-skew tolerance, invocation tracing, and durable events logging; deployed to **~1,322** apps.
- Contributed to ACA Express Public Preview on Azure Dev Compute (ADC), resolving P0 blockers for ephemeral volumes and Log Analytics customer logs; unblocked **~100K+** ACA environments.
- Implemented ACA Sessions management APIs and fixed critical async execution bugs for **~300** external customers; enabled early Fabric session deletion to reduce resource cost.
- Led Dapr AKS Extension rollouts (v1.15, v1.16), remediating CVEs across **~400** AKS clusters and integrating CodeQL scanning for enhanced security posture.
- Integrated ACA agentic apps with Foundry Control Plane (FCP), supporting declared and auto-detected agents and central management for **~2,000+** applications.

Software Development Engineer I

Jul 2021 – Jul 2024

Microsoft – Dapr

Hyderabad, India

- Stabilized Dapr building blocks across Configuration API, MQTT PubSub, CosmosDB state and Redis/PostgreSQL config stores. Added support for Event Hubs bulk-subscribe and CLI airgap install.
- Implemented Cascade Terminate/Purge for Dapr Workflows and fixed scalability bottlenecks found through performance tests.
- Co-launched ACA Sessions, owning Node.js/.NET code-interpreter support for Security Copilot; fixed critical bugs and added regression coverage for **3000+** active sessionPools.
- Designed patch-based internal fork infrastructure across **5** Dapr repositories; built OneBranch/Ev2 pipelines and automated OSS-to-internal releases with **<1** day SLA.
- Delivered multi-architecture Dapr/dashboard images, 1P Container Registry migration, ACA nightly builds, and Mariner/distroless images to MCR.

Quantitative Research Intern

May 2020 – Jul 2020

JP Morgan & Chase – E-Trading

Mumbai, India (Remote)

- Built price prediction models using **LSTM**, feedforward neural networks, and regression with univariate feature selection; improved predictive power by **40%** for HK names through feature standardization.

RESEARCH

Data Race Detection for Task-Parallel Programs

Jun 2019 – Aug 2021

Supervisor: Prof. Swarnendu Biswas, IIT Kanpur

- Designed **Tasker**, combining vector clocks with tree-based techniques; built an **LLVM** memory instrumentation pass and achieved **1.48x speedup** over PTRacer with lower memory overhead.
- “Efficient Data Race Detection of Structured Task Parallel Programs Using Vector Clocks” — S. Kumar, A. Agrawal, S. Biswas | Submitted to ISCGO 2022.

PROJECTS

GemOS | C, x86 – OS kernel (IIT Kanpur CS330)

Aug 2019 – Nov 2019

Implemented mmap/munmap/mprotect with lazy allocation and page-fault handling; cfork/vfork with copy-on-write semantics.

Cipher Decoder | Python, Modern Cryptology (CS641)

Aug 2020 – Nov 2020

Performed differential cryptanalysis of 3-DES and decrypted Caesar/Vigenere ciphers using frequency analysis.

EDUCATION

Indian Institute of Technology, Kanpur

CGPA: 9.4/10.0

Bachelor of Technology in Computer Science and Engineering

2017 – 2021

TECHNICAL SKILLS

Languages:

Go, C#/.NET, Python, C/C++

Cloud & Infra:

Azure (ACA, Functions, AKS), Kubernetes, Docker, Helm, KEDA

Distributed Systems:

Dapr (CNCF), Microservices, Service Mesh, gRPC, Pub/Sub

ACHIEVEMENTS & RECOGNITION

- IDC Pinnacle FY'26 – Best AI Success Story** (ACA RCA Agent squad, ~200+ competing teams); **RockStar Award**, Microsoft ('22, '23).
- Academic Excellence Award** (x2), IIT Kanpur | JEE Adv. **AIR 715** | JEE Main **AIR 348** | KVPY Scholar | NSE Physics Top 1%.